

1. Urban Mobility

Problem Statement : Enhancing Urban Commuting Efficiency and Non-Motorized Road User Safety

Rapid urbanization has intensified challenges in city transport systems, leading to severe bottlenecks in everyday mobility and rising safety concerns for non-motorized road users.

Commuters in dense urban centers face major inefficiencies during daily travel, spending excess time and fuel navigating congested corridors and searching for available parking spaces. The lack of integrated, real-time mobility guidance prevents drivers from choosing optimal routes and parking solutions dynamically, worsening gridlock and increasing urban emissions.

Concurrently, pedestrians and cyclists continue to experience high risks of conflict with vehicular traffic, particularly at unsignalled intersections, poorly illuminated stretches, and high-density crossing zones. Traditional static infrastructure often fails to provide timely warnings to drivers or vulnerable road users when safety hazards arise.

Keywords : Public transport, traffic, parking, ride-sharing, last-mile delivery, pedestrian safety, etc

2. Smart Commerce

Problem Statement : Empowering Small Retailers Through Digital Visibility and Automated Store Operations

Small and independent brick-and-mortar businesses face significant operational and competitive hurdles in an increasingly digitalized economy. While large enterprise retailers leverage automated systems and online ecosystems, local shop owners remain constrained by limited reach and manual, error-prone store management practices.

Nearby shoppers frequently lack real-time visibility into local product availability, leading them to rely on distant e-commerce platforms rather than supporting nearby merchants. This disconnect reduces foot traffic and revenue potential for neighbourhood stores. Simultaneously, inside the physical store, manual checkout processes slow down transaction times, lead to long wait times for customers, and expose retailers to inventory discrepancies and unrecorded shrinkage due to limited oversight.

Keywords : Retail, shopping, small businesses, payments, inventory, customer experience, local sellers, etc

3. Climate & Natural Resources

Problem Statement : Optimizing Local Resource Management through Continuous Monitoring and Behavioral Insights

Residential spaces, educational campuses, and small enterprises lack real-time visibility into their daily resource consumption—specifically across water, energy, and waste streams. Without actionable, centralized data, individuals and facility managers struggle to identify inefficiencies, track usage patterns, or adopt targeted conservation practices, leading to avoidable resource depletion and elevated operational costs.

Unnoticed infrastructure failures, particularly hidden water leaks across supply pipelines and distributed plumbing networks, cause severe, cumulative resource loss before manual detection occurs. Traditional static monitoring methods fail to identify abnormal flow rates or notify stakeholders dynamically, allowing leaks to persist unchecked and compounding structural and environmental damage.

Keywords : Water, energy, waste, forests, pollution, extreme weather, conservation, resource management, etc.

4. Connected Communities

Problem Statement : Accelerating Emergency Response and Enhancing Safety Accessibility Across Communities

During critical safety incidents and localized emergencies, residents in residential neighborhoods and public spaces face severe friction in initiating timely calls for help and obtaining actionable situational updates. Traditional reporting channels are often fragmented, causing delays in dispatching local authorities, coordinating nearby volunteers, and providing clear status updates to those affected.

The issue is further exacerbated for vulnerable demographics—such as children, senior citizens, and individuals with physical or cognitive disabilities—who may find navigating complex smartphone applications difficult or impossible during high-stress scenarios. When an emergency strikes, relying exclusively on app-based reporting excludes these groups from receiving rapid, accessible assistance.

Keywords : Neighborhoods, public services, local governance, emergency response, community engagement, accessibility, etc.

5. Travel & Exploration

Problem Statement : Enhancing Personalized Cultural Exploration and Belonging Security for Travelers

Travelers visiting unfamiliar cities frequently struggle to curate cohesive, time-efficient itineraries that align with their specific personal interests. As a result, mainstream travel routes often overshadow lesser-known cultural, historical, and local heritage sites, leaving visitors with generic travel experiences and minimal contextual understanding of the destinations they explore.

At the same time, journey security remains a primary concern for both solo and group travelers. Mishandled luggage, misplaced personal belongings, and fear of property loss introduce significant stress and disruption to travel schedules. Relying on passive transit tracking or traditional manual checks leaves travelers vulnerable to property displacement without immediate awareness or dynamic location recovery options.

Keywords : Tourism, navigation, hotels, cultural heritage, travel planning, local experiences, accessibility, safety, etc.

6. Entertainment & Creative World

Problem Statement 1: Streamlining Creator Workflows and Elevating In-Person Audience Engagement

Independent content creators across digital art, video, and audio domains are often overwhelmed by technical, administrative, and post-production tasks. Spending excessive effort on scripting, editing, and manual performance analysis diverts critical time and focus away from original artistic creation, limiting content output and potential growth.

Concurrently, live entertainment venues—such as concerts, sports screenings, and university events—struggle to deliver dynamic, highly interactive experiences that fully engage modern audiences. Traditional event setups rely on static or centrally controlled environmental effects that fail to adapt responsively to real-time crowd energy, music variations, or physical participation, leaving live experiences feeling passive.

Keywords : Movies, music, gaming, content creation, sports entertainment, digital art, creators, events, etc.

7. Bioinformatics

Problem Statement 1 : Accelerating Multi-Omics Data Integration and Predictive Biomarker Discovery for Precision Medicine

Modern biomedical research generates massive volumes of heterogeneous multi-omics data—including genomics, transcriptomics, proteomics, and metabolomics—across disparate repositories and formats. Researchers and clinical scientists struggle to effectively integrate these high-dimensional datasets to uncover meaningful biological insights due to high computational overhead, fragmented analysis pipelines, and a lack of unified, real-time visualization frameworks.

Concurrently, identifying reliable disease-specific diagnostic or prognostic biomarkers remains a slow, error-prone manual process. Traditional computational models often suffer from overfitting and struggle to handle missing clinical or molecular variables, delaying the translation of raw sequencing data into actionable therapeutic targets or personalized treatment strategies.

Keywords: Genomics, proteomics, multi-omics, biomarker discovery, precision medicine, computational biology, drug discovery, biological networks, etc.

8 . Biotechnology

Problem Statement 1 : Optimizing Bioprocess Control and Real-Time Metabolic Monitoring for Sustainable Bio-Production

Industrial and laboratory-scale biomanufacturing processes—such as microbial fermentation, enzymatic synthesis, and cell culture growth—frequently suffer from suboptimal yield efficiency and batch-to-batch variability. Bio-engineers and operators lack intuitive, centralized systems for continuous, real-time monitoring of critical process parameters like dissolved oxygen, pH, nutrient consumption rates, and biomass accumulation. Without dynamic feedback control loops and automated anomaly detection, unexpected metabolic shifts or environmental fluctuations can lead to complete batch failures, prolonged production cycles, and excessive resource waste. Traditional static sampling methods fail to provide timely alerts, leaving operators unable to execute corrective interventions before cellular productivity or product quality is severely compromised.

Keywords : Bioprocessing, fermentation, metabolic engineering, bioreactors, synthetic biology, enzyme kinetics, bio-manufacturing, process optimization, etc.

9 . Cybersecurity

Problem Statement 1 : Mitigating Advanced Digital Threats and Securing Distributed Infrastructure

As organizations, critical public utilities, and digital systems become increasingly interconnected, the attack surface for sophisticated cyber threats continues to expand exponentially. Enterprises and public sector entities struggle to maintain real-time visibility and defensive posture against advanced persistent threats (APTs), automated ransomware vectors, and complex supply chain vulnerabilities. Traditional reactive security frameworks often fail to detect novel attack patterns or respond swiftly enough to mitigate zero-day exploits before critical data or infrastructure is compromised. Furthermore, resource-constrained teams face significant friction in automating threat intelligence sharing, enforcing zero-trust access controls, and maintaining continuous compliance across distributed cloud and edge environments.

Keywords : Threat intelligence, zero-trust architecture, encryption, intrusion detection, vulnerability assessment, incident response, network security, blockchain security, etc .

10 . Open Innovation & Cross-Departmental Solutions

Problem Statement 1 : Harnessing Cross-Disciplinary Synergies for Multi-Departmental Problem-Solving and Resource Harmonization

Complex societal, industrial, and institutional challenges frequently span multiple operational boundaries—such as civil infrastructure, mechanical automation, business management, and socio-economic planning—requiring seamless integration across diverse academic and technical fields. However, teams working on cross-functional initiatives face severe friction in sharing domain-specific data, standardizing cross-departmental metrics, and synthesizing cohesive collaborative frameworks.

The lack of interoperable collaboration tools and unified evaluation environments leads to fragmented problem-solving, redundant efforts, and misaligned strategic outcomes when combining hardware, software, economic, and administrative insights. Without automated cross-domain correlation mechanisms, stakeholders struggle to evaluate systemic trade-offs or implement holistic, multi-faceted solutions effectively across diverse engineering and non-engineering departments.

Keywords : Cross-functional collaboration, systems thinking, socio-technical systems, multi-departmental innovation, integrated design, institutional efficiency, complex systems, open innovation, etc. .

